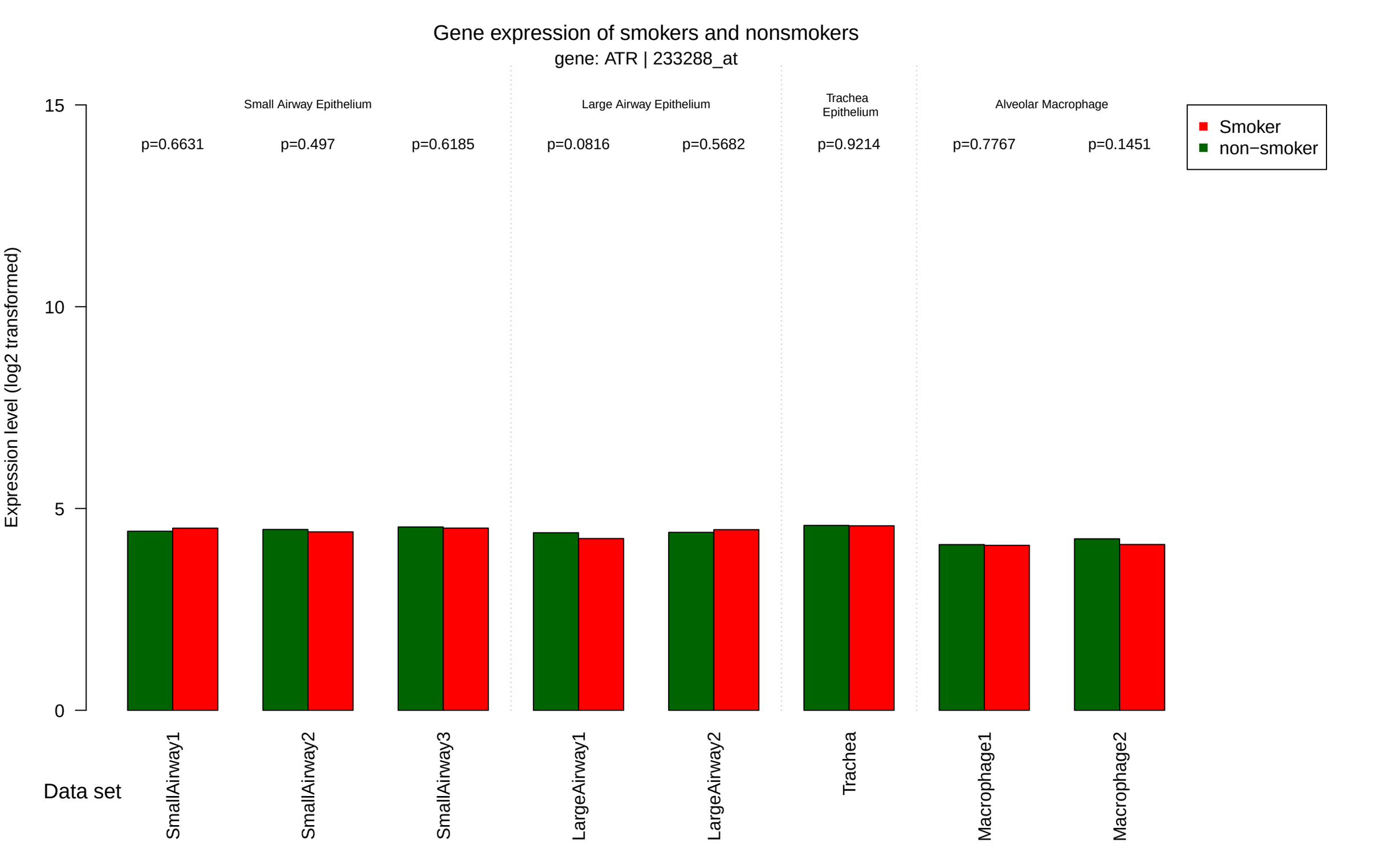
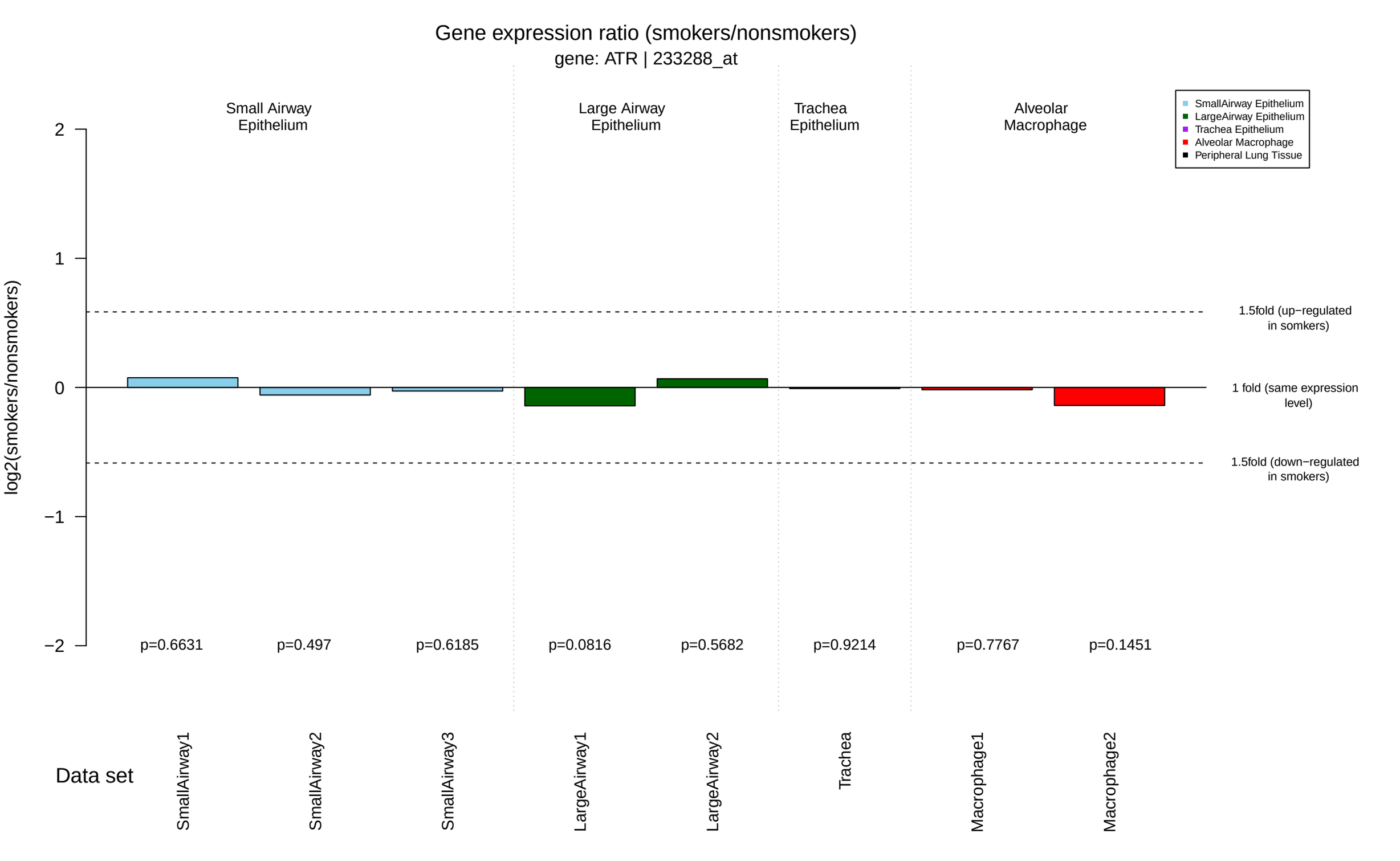
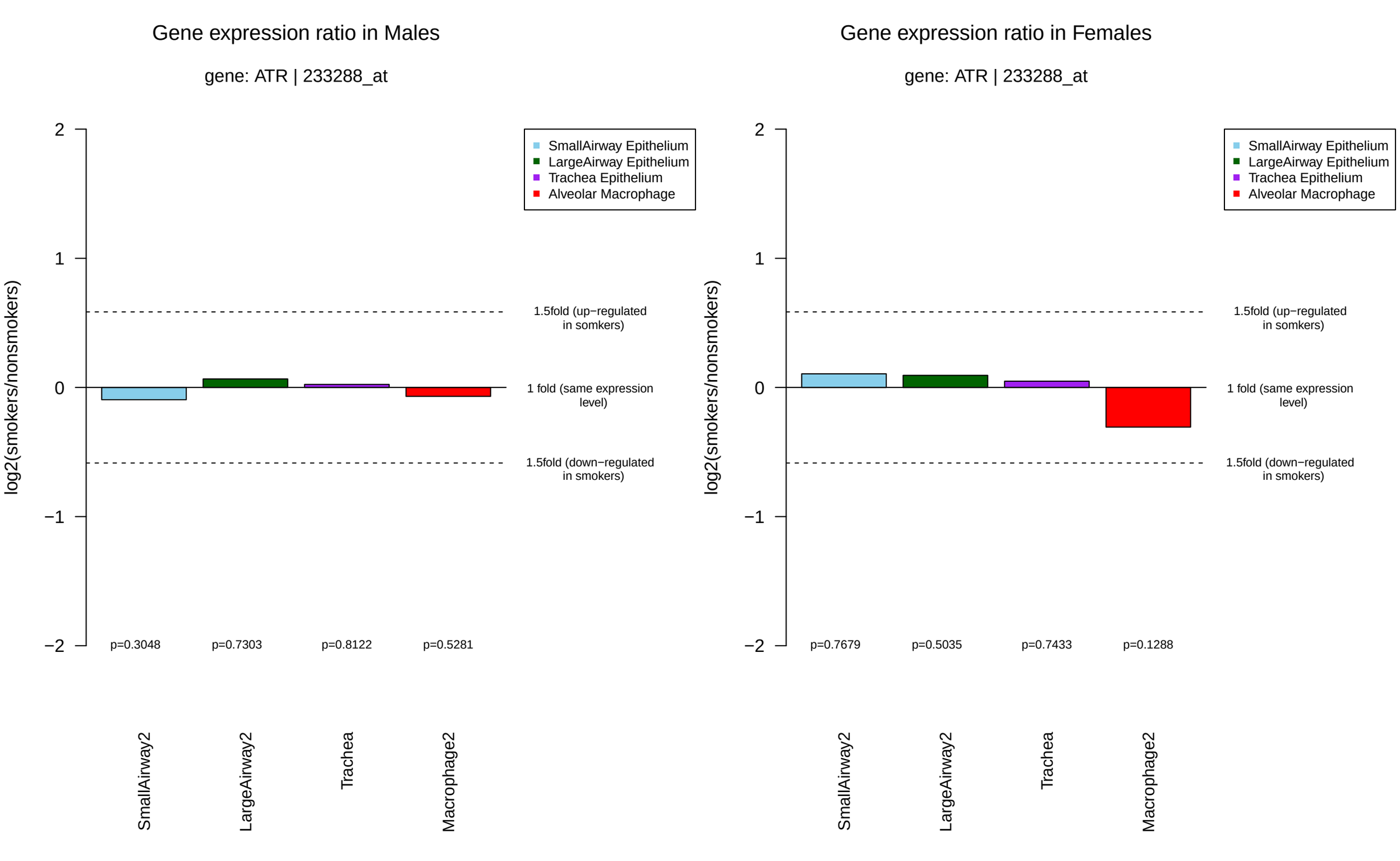
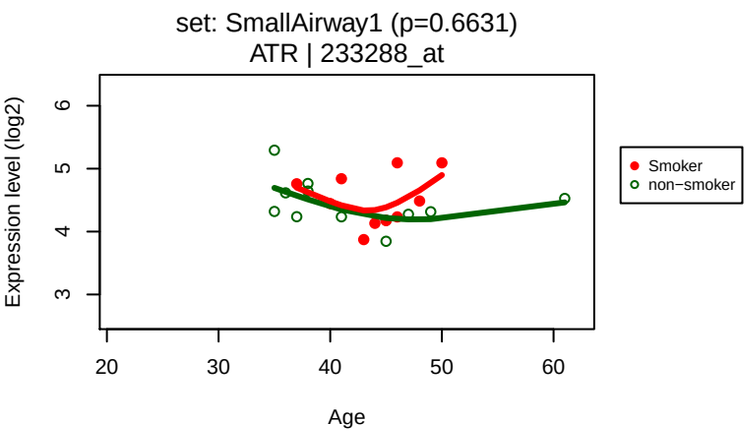
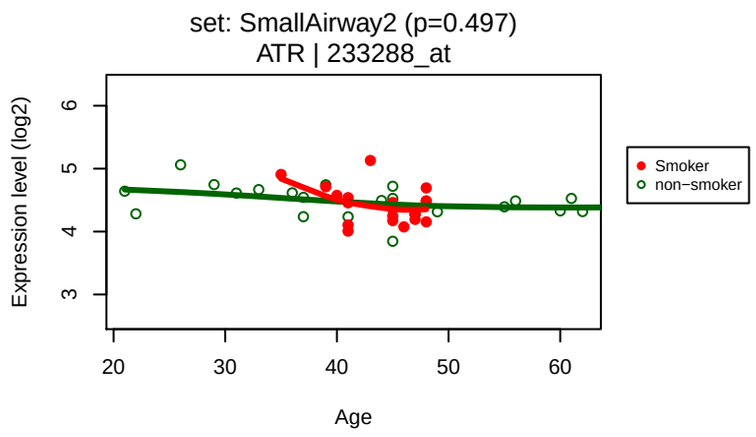
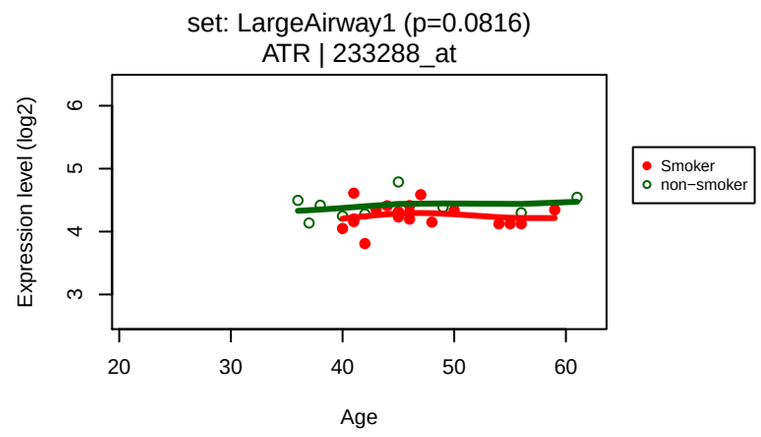

Gene expression of smokers and nonsmokers
gene: ATR | 233288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6631
p=0.497
p=0.6185
p=0.0816
p=0.5682
p=0.9214
p=0.7767
p=0.1451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

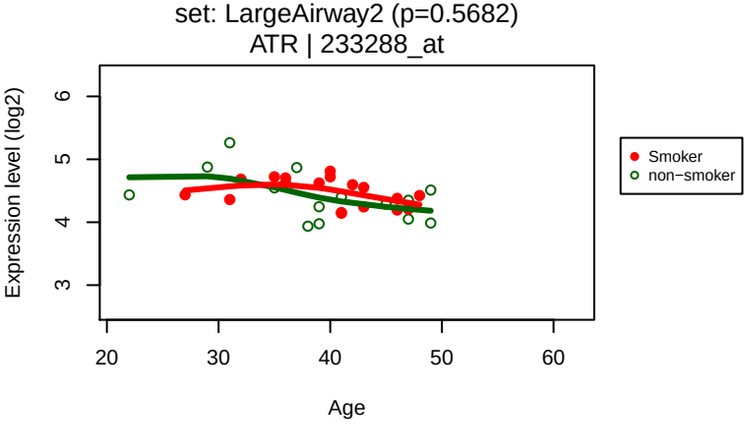

Gene expression ratio (smokers/nonsmokers)
gene: ATR | 233288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6631
p=0.497
p=0.6185
p=0.0816
p=0.5682
p=0.9214
p=0.7767
p=0.1451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

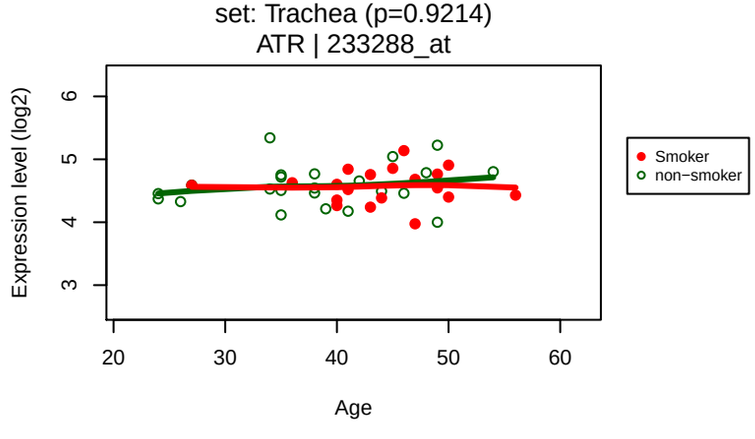

Gene expression ratio in Males
gene: ATR | 233288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3048
p=0.7303
p=0.8122
p=0.5281
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATR | 233288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7679
p=0.5035
p=0.7433
p=0.1288
SmallAirway2
LargeAirway2
Trachea
Macrophage2

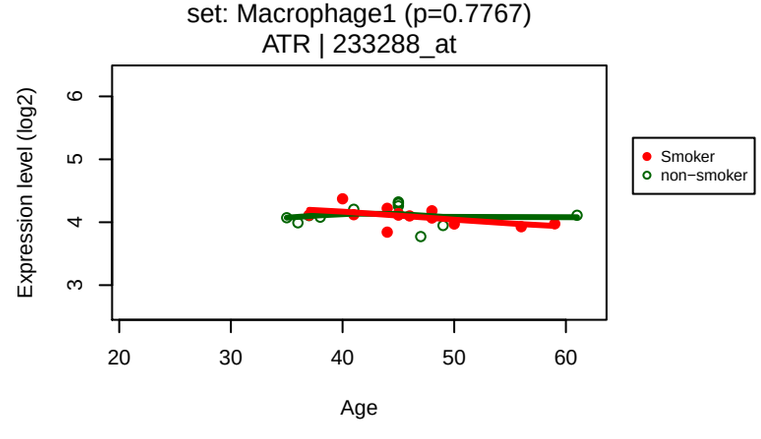

set: SmallAirway1 (p=0.6631)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker

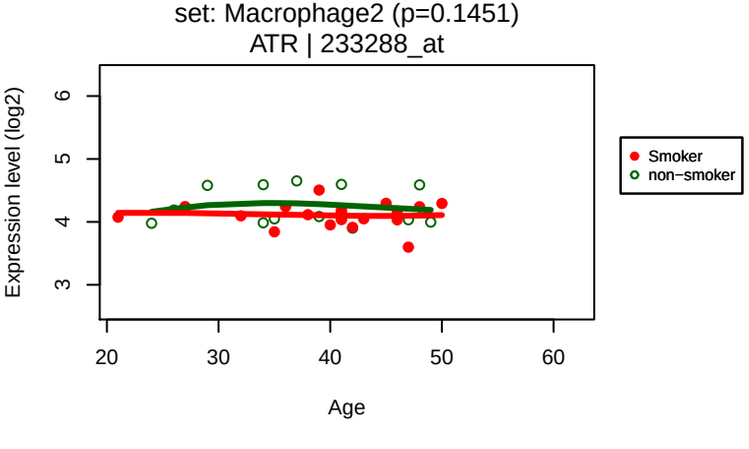

set: SmallAirway2 (p=0.497)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker

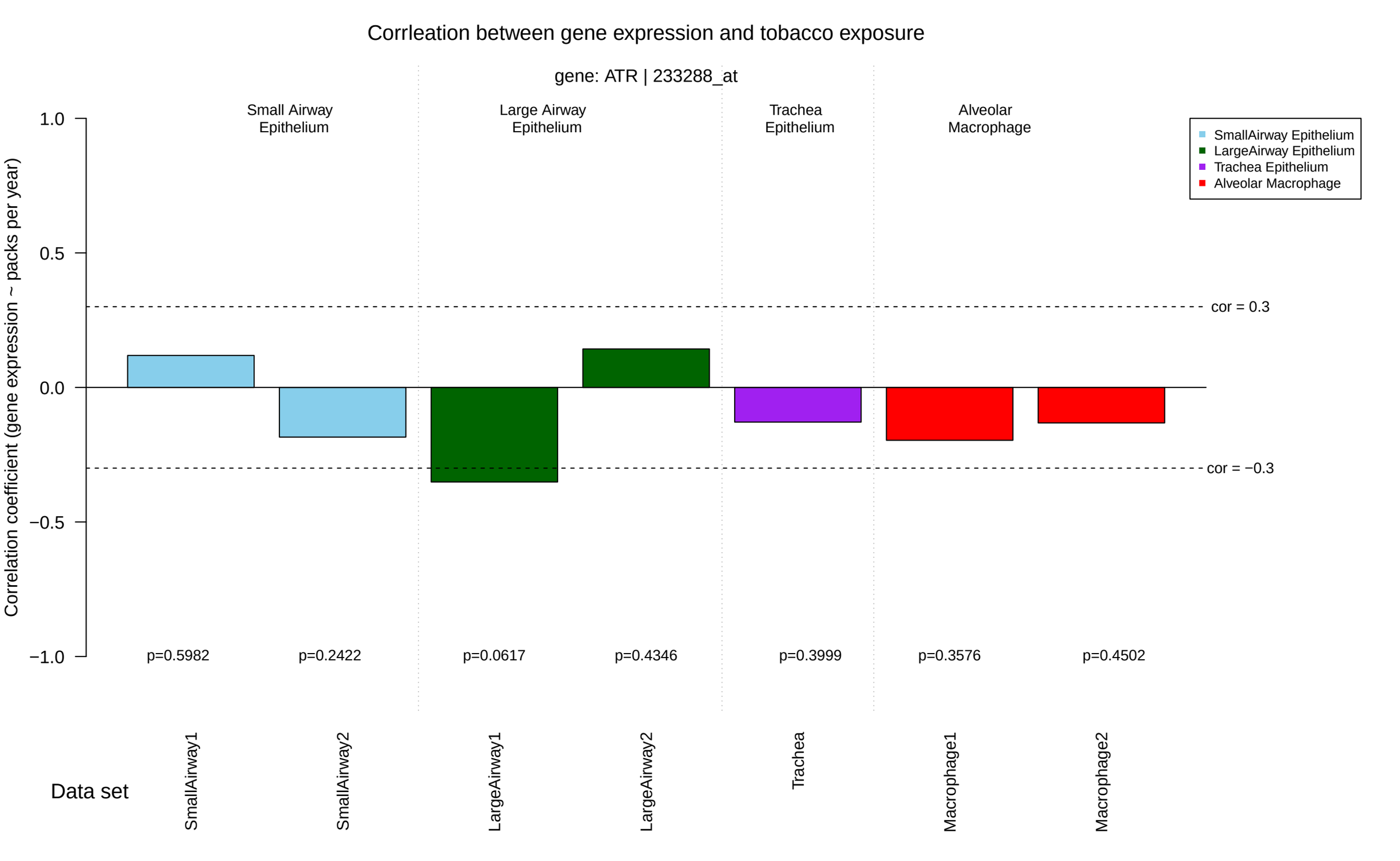

set: LargeAirway1 (p=0.0816)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5682)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9214)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7767)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1451)
ATR | 233288_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATR | 233288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5982
p=0.2422
p=0.0617
p=0.4346
p=0.3999
p=0.3576
p=0.4502
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2